Intelligent Electricity Billing System with Consumption analytics

Report : Intelligent Electricity Billing System.

Problem Definition : Develop a comprehensive electric billing system that calculates bills, analyzes consumption patterns, and implements smart pricing logic using computational ~~logi~~ thinking principles. Apply logical reasoning. Conditional programming. & systematic problem solving approaches to solve the proposed work.

Taskwise Explanation

Task 1: Basic Billing Calculation.

This task calculates the total electricity bill based on the number of units consumed.

It uses simple conditions.

- No charge for the first 100 units.
- ₹5 per unit for next 100 units (101-200)
- ₹10 per unit for unit above 200

Finally it displays the total units & bill amount

Task 2: Bill breakdown and Analytics.

This task gives a detailed breakdown of the bill for each slab (0-100, 101-200, and above 200). It also checks if the input is valid and then shows how much the user spent in each slab. Based on total units. it clasifies users as.

- Low consumtion (ECO-friendly)
- Average consumption.
- High consumption (Energy alert)

Task 3: Discount and Recommendations.

This task adds discount and energy saving advice If the user consumes more than 200 units, a 10% discount is given. It also provides personalized recommendations

- Low users get encouragement.
- Medium users get simple saving tips.
- High users get advice to reduce heavy usage.

Task 4: Comparison Between Two users.

This task compares electricity usage and bills between two users it shows each users total units and bills, then tells who paid more and by what percentage, or if both paid the same. It helps users understand their consumption compared to others.

Task 5: Optional - Advanced Features.

This task adds smart billing features.

- Seasonal pricing: Bills change w/ the season (higher in summer lower in winter).
- Peak hour pricing: Higher rates during busy hours
- forecasting: predicts next month's usage based on the last three months.

Computational Thinking.

1. Decomposition.

In the catering management system, the large task of manging catering services is divided into smallar parts such as menu management, order processing, billing, inventory control, and customer feed back. This helps make the problem easier to handle and allows each part to be designed and improved seperately.

2. Pattern Recognition:

The system identifies common patterns such as frequently ordered dishes, peak hours for orders, and repeated customer preferences. Recognizing these patterns helps the business plan ingredients, reduce waste, and provide better customer service.

3. Abstraction

It focuses on only on the important details like item names, price quantities, and total cost while ignoring unnesessary information. For example, Instead of worrying about every step of food preparation the system records the total items and costs fo billing.

4. Algorithm Design:

The system follows a step by step process to processing tasks such as taking an order, calculating the total bill, applying discounts, and generating reciepts.

Each problem is logical and easy to follow, ensuring accurate and consistent results every time.

Algorithm characteristics

1. Input : units (numeric)
2. output: bills amount, breakdown cutergories, comparisons.
3. Definiteness: Defenite steps for slab calculations
4. Finiteness: Executes finites arithmatic and returns result.
5. Effectiveness: Uses simple arithmatic / conditional, implementable in codes.
6. Greenerally : works for any non-ve consumption value.

Conclusion: The provided codebase implements the requested tasks, generates sample outputs and provides an extensible base for advance features.